LEVEL 26 — Polynomial Operations

Easy NumPy polynomial examples with line-by-line code and output.

1. Polynomial Creation

Explanation: A polynomial is represented by its coefficients from highest power to lowest power.

Code:

```
import numpy as np

# 2x² + 3x + 1
p = np.poly1d([2, 3, 1])

print(p)
```

Output:

```
   2
2 x + 3 x + 1
```

Important: [2, 3, 1] represents $2x^2 + 3x + 1$.

2. Polynomial Evaluation

Explanation: Polynomial evaluation means finding the value of a polynomial for a given x.

Code:

```
import numpy as np

p = np.poly1d([2, 3, 1])

x = 2
result = p(x)

print(result)
```

Output:

```
15
```

Important: $2(2^2) + 3(2) + 1 = 8 + 6 + 1 = 15$.

3. Polynomial Fitting

Explanation: Polynomial fitting finds a polynomial that best fits given x and y data.

Code:

```
import numpy as np

x = np.array([1, 2, 3])
y = np.array([2, 4, 6])

coefficients = np.polyfit(x, y, 1)

print(coefficients)
```

Output:

```
[2.00000000e+00 1.30728761e-15]
```

Important: Degree 1 means a straight-line fit. The result is approximately $y = 2x$.

4. np.polyval()

Explanation: np.polyval() evaluates a polynomial at given x values.

Code:

```
import numpy as np

coefficients = [2, 3, 1]

x = 2
result = np.polyval(coefficients, x)

print(result)
```

Output:

```
15
```

Important: The polynomial is $2x^2 + 3x + 1$. At x=2, the answer is 15.

5. np.polyfit()

Explanation: np.polyfit() finds polynomial coefficients that fit x and y data.

Code:

```
import numpy as np

x = np.array([1, 2, 3, 4])
y = np.array([1, 4, 9, 16])

coefficients = np.polyfit(x, y, 2)

print(np.round(coefficients, 2))
```

Output:

```
[ 1.  0. -0.]
```

Important: Degree 2 fits $y = x^2$, so the coefficients are approximately [1, 0, 0].

6. Polynomial Roots

Explanation: Roots are the x values where the polynomial becomes zero.

Code:

```
import numpy as np

# x² - 5x + 6
p = np.poly1d([1, -5, 6])

print(p)
print(np.roots(p))
```

Output:

```
   2
1 x - 5 x + 6
[3. 2.]
```

Important: The roots are 3 and 2 because (x-3)(x-2)=0.

7. np.roots()

Explanation: np.roots() finds the roots of a polynomial from its coefficients.

Code:

```
import numpy as np

coefficients = [1, -5, 6]

roots = np.roots(coefficients)

print(roots)
```

Output:

```
[3. 2.]
```

Important: For $x^2 - 5x + 6 = 0$, the solutions are $x=3$ and $x=2$.

8. Polynomial Derivative

Explanation: A derivative gives the rate of change of a polynomial.

Code:

```
import numpy as np

p = np.poly1d([3, 2, 1])

derivative = np.polyder(p)

print(derivative)
```

Output:

```
3 x + 2
```

Important: For $3x^2 + 2x + 1$, the derivative is $6x + 2$.

9. Polynomial Integration

Explanation: Integration finds an antiderivative of a polynomial.

Code:

```
import numpy as np

p = np.poly1d([3, 2, 1])

integral = np.polyint(p)

print(integral)
```

Output:

```
     3     2
1.0 x + 1.0 x + 1.0 x + C
```

Important: The integral of $3x^2 + 2x + 1$ is $x^3 + x^2 + x + C$.

Quick Revision

• np.poly1d() → creates a polynomial

- Polynomial evaluation → finds the polynomial value at x
- np.polyfit() → fits a polynomial to data
- np.polyval() → evaluates polynomial coefficients
- np.roots() → finds polynomial roots
- np.polyder() → calculates the derivative
- np.polyint() → calculates the integral